

LESSON GUIDE

Technology Plus

Unit 7: Security

6.2.4 – Safe Browsing Practices

Lesson Overview:

In this lesson, students will learn safe browsing practices and compare valid and invalid web certificates.

Students will:

- Explain what a digital certificate is and why it matters for safe browsing.
- Differentiate between a valid and invalid certificate.
- Compare valid and invalid certificates from a web browser

Guiding Question: How can you tell if a website is safe to use, and why it is important to check for a valid certificate when browsing online?

Suggested Grade Levels: 8-12

Technology Needed: No

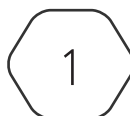
Approximate Time Required: 50-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.2 – Explain methods to secure devices and security best practices.
 - Safe browsing practices
 - Certificates
 - Valid

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Lesson Background

Background Information:

When browsing the internet, users often come across websites that request sensitive information such as login credentials, personal details, or payment data. To ensure that communication between the user and the website is secure, many websites use digital certificates. These certificates are issued by trusted organizations called Certificate Authorities (CAs) and serve as proof that the website is legitimate and that the data exchanged will be encrypted.

A valid certificate means: The certificate was issued by a trusted CA. It has not expired or been revoked. It matches the website's domain name. When certificates are valid, users can trust that their connection is encrypted (HTTPS) and that their information is protected from attackers who might try to intercept or impersonate the website. Browsers typically show a padlock icon next to the web address to indicate a valid, secure certificate. If a certificate is invalid, users may see warnings like "Your connection is not private" or "Certificate expired/invalid." Safe browsing practices mean avoiding websites with invalid certificates, as they could expose data to attackers or be phishing attempts.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.2.4 – Safe Browsing Practices • Activity Slides 6.2.4 – Certificate Analysis
Materials per Student	• Student Note 6.2.4 – Safe Browsing Practices • Student Handout 6.2.4 – Safe Browsing Practices • Assessment 6.2.4 – Safe Browsing Practices (optional)

Cyber Terms:

Term	Definition
Certificate Authority (CA)	a trusted organization that issues certificates

6.2.4 – Safe Browsing Practices

Guiding Question: How can you tell if a website is safe to use, and why is it important to check for a valid certificate when browsing online?

Lesson Launch (5 minutes)

- **Ask:** You are about to buy something online. A warning pops up saying the site's certification is invalid. Do you continue?
- What do you think a certificate is?
- How do you know if a website is safe to use?
- Allow time between each question for student discussion

Safe Browsing (20-25 minutes)

- Distribute the Student Handout 6.2.4 – Safe Browsing Practices and have students fill in the answers during the lesson.
- Discuss safe browsing practices.
 - Use Secure Websites (HTTPS)
 - Look for HTTPS in the web search bar. HTTPS encrypts the data exchanged between your browser and the website, making it difficult for attackers to intercept data.
 - Look for the padlock icon in the web search bar. Most browsers display a padlock to indicate a secure connection although, the padlock icon is not a foolproof sign of a legitimate site.
 - Be Cautious with Links and Attachments.
 - Verify links by hovering over them to preview their destination before clicking. Avoid suspicious or unsolicited links, especially from unknown sources.
 - Beware of Attachments: Don't download or open attachments from unfamiliar or untrusted sources.
 - Use Strong, Unique Passwords
 - Create strong, complex passwords with a mix of numbers, letters, and symbols without any personal information.
 - Use a password manager to generate and store strong, unique passwords for each account you have.
 - Enable Multi-Factor Authentication (MFA)
 - Using MFA adds an extra layer of security and involves using two or more factors of authentication, something you know, something you have, and something you are.
 - Keep Your Browser Updated

- Install updates regularly to get the latest security patches and improvements.
- Update extensions or add-ons often and only download them from reputable sources.
- Avoid Public Wi-Fi
 - Use virtual private networks (VPNs) in public especially when accessing sensitive information or conducting financial transactions.
- Be Aware of Phishing Attempts
 - Be cautious of emails, messages, or websites that ask for personal information or login credentials.
- Manage Privacy Settings
 - Configure your browser's privacy settings to limit tracking and data collection by third-party websites and advertisers.
 - Regularly clear cookies and cache to remove tracking data and protect your privacy.
- Discuss web certificates.
 - Introduce a **certificate authority (CA)**
 - There are multiple trusted CAs.
 - Discuss types of certificates
 - In this lesson we are only looking at web browsing certificates.
 - Share more tips:
 - Certificates are the backbone of HTTPS.
 - Certificates follow a chain of trust; they don't stand alone. Web browsers check the root certificate, intermediate certificate, and server certificate before displaying the padlock on a web address.
 - Root: trusted by the browser/OS
 - Intermediate: issued by the CA to help secure distribution
 - Server: on the website a user is currently visiting
 - Sometimes certificates are revoked before they expire, if the private key is stolen or the site is hacked.
 - Some websites have self-signed certificates that have not been issued by a trusted CA.
 - Your browser/OS keeps a built-in list of trusted CAs.
 - Even malicious sites can have valid certificates because DV certificates are cheap and easy to get. A padlock = encryption, but don't guarantee the site is legitimate.
- Discuss valid vs. invalid certificates.
- **Did You Know:** DigiNotar (short for Digital Notary) was a Dutch company that issued certificates. In 2011, hackers broke into their systems and created fake certificates. Once the breach was discovered, major browsers like Chrome and Firefox removed DigiNotar from their list of trusted Certificate Authorities. DigiNotar technically kept issuing certificates, but because browsers no longer recognized

them, those certificates weren't accepted by most users' devices. Once DigiNotar lost trust, it could never regain it, and the company went bankrupt within weeks. This real-world case shows why CA security is critical and why browsers maintain their own trust lists.

Activity: Certificate Analysis (20-25 minutes)

- Display Activity Slides 6.2.4 – Certificate Analysis, have students record their answers in the handout.
- As you show each certificate screenshot, have students fill in the handout table with:
 - Website URL
 - Issued To
 - Issued By
 - Valid Dates
 - Valid or Invalid status
- Point out that while the last three certificates use the same URL, differences in the issuing authority, dates, and trust status make some valid and others invalid.
- After completing the table, review the discussion questions with the class.

Certificate Analysis Table

Website URL	Issued To	Issued By	Valid Dates	Valid or Invalid?
www.apple.com	Apple	Apple Inc.	03/04/25 -10/16/25	Valid
ssls.com	SSLS	Sectigo Limited	09/10/24 -10/03/25	Valid
github.com	Github	Sectigo Limited	02/04/25 - 02/05/26	Valid
badssl.com	BadSSL	COMODO CA Limited	04/08/15 -04/12/15	Invalid
badssl.com	BadSSL	BadSSL	08/26/25 -08/26/27	Invalid
badssl.com	BadSSL	BadSSL Untrusted Root	08/26/25 -08/26/27	Invalid

Activity Debrief Questions and Answers

1. How can you tell a valid certificate from an invalid one?

A valid certificate is issued by a trusted CA, matches the domain name, has valid dates, and has not been revoked.

2. Why is it important to check certificates before entering sensitive info?

Certificates validate if a site is safe for entering sensitive info by confirming the connection between the browser and the server are secure and cannot be intercepted by an attacker.

Putting It All Together (5-10 minutes)

- Review what students learn today by discussing the questions below as a class?
- What are some safe browsing practices you already follow?

Student answers will vary but should include something from the safe browsing practices list.

- How can we tell the difference between a valid and invalid certificate?

Valid web certificates are issued by trusted CAs, match the website's domain name, have a valid date range, and are not revoked. Invalid certificates are issued by an untrusted CA or self-signed certificate, the domain name doesn't match the site, or it has expired or been revoked.

- Why should we make sure a website has a valid certificate before we visit it?

Certificates validate if a site is safe for entering sensitive info by confirming the connection between the browser and the server are secure and cannot be intercepted by an attacker.